FPGA-Based Traffic Light Controller with Priority System

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Internship

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Project Overview

This project implements a Traffic Light Controller using Verilog HDL. The controller manages traffic signals through Red, Yellow, and Green states. An emergency vehicle priority system is integrated to provide immediate Green signal access whenever an emergency vehicle is detected.

Objectives

- Design a traffic light controller using Verilog HDL.
- Implement a Finite State Machine (FSM) for signal control.
- Provide emergency vehicle priority override.
- Demonstrate FPGA-compatible digital logic design.

Features

- Automatic traffic signal sequencing.
- Emergency vehicle priority control.
- Finite State Machine (FSM) implementation.
- FPGA-compatible Verilog design.
- Simple and scalable architecture.

Files Included

- traffic_light_controller.v
- testbench.v
- README.md
- Project_Report.pdf

Working Principle

Normal Operation

1. Red Light ON
2. Green Light ON
3. Yellow Light ON

4. Repeat cycle

Emergency Mode

1. Emergency signal detected.
2. Controller immediately switches to Green.
3. Traffic flow is prioritized for emergency vehicles.
4. System returns to normal operation after emergency clearance.

Applications

- Smart Cities
- Intelligent Transportation Systems
- Emergency Vehicle Routing
- Urban Traffic Management

Tools Used

- Verilog HDL
- FPGA Design Methodology
- Digital Logic Design

Conclusion

The Traffic Light Controller with Priority System successfully demonstrates traffic management using Verilog HDL. The emergency override feature improves road safety and emergency response efficiency.